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RESEARCH INTERESTS

Medical Image Processing, Biomedical Signal Processing, Generative AI, Large Language Models (LLMs), Domain Adaptation, Disease Detection, Bioimpedance, Health Informatics, Neuroengineering.

EDUCATION

University of Dhaka

Master of Science in Biomedical Physics and Technology

Dhaka, Bangladesh

10 May, 2026 – June 2027 (Expected)

- **Specialization:** Biomedical Engineering

Khulna University of Engineering and Technology

Bachelor of Science in Biomedical Engineering

Khulna, Bangladesh

11 November, 2018 – 27 February, 2024

- **CGPA: 3.77/4.00** (80.60%) [Junior/Senior GPA: 3.80] (Ranked 4th in overall merit position)
- Evaluated by **Educational Credential Evaluators (ECE)** on December 30, 2024
U.S. Equivalence: Bachelor Degree; Major Area of Study: Biomedical Engineering, **Grade Average: 3.91/4.00**
- **UG Thesis:** “Gallbladder Cancer Classification using Parallel Transfer Learning with Multi-model Feature Fusion and LSTM”
Summary: Developed a hybrid framework combining parallel transfer learning, multi-model feature fusion, and LSTM for gallbladder cancer classification, achieving 99.37% accuracy.

RESEARCH EXPERIENCE

Researcher

Bio-Neural Intelligence and Research Advancement Lab, KUET

June 2025 – Current

Khulna, Bangladesh

- Designed NeuroSwin, a hybrid deep learning model (Swin Transformer-GRU), to classify Parkinson’s Disease and identify its source regions in the brain from EEG signals.

Research Assistant Intern

Ulster University

October 2024 – Current

Londonderry, Northern Ireland, UK

- Developed **TransfusionNet**, a hybrid architecture augmented with a specialized preprocessing pipeline and an early-layer feature fusion paradigm for cervical cancer detection. This work has been published in the **Q1** journal **Results in Engineering** (Elsevier, Impact Factor: 7.9).
- Created a benchmark to evaluate the performance of large language models (LLMs) in identifying disease classes, comparing results against our proposed architecture MobileCoAtNet for stomach infection classification.
- Investigated the adaptability of foundational AI models for multi-cancer classification, implementing Model-Agnostic Meta-Learning (MAML) for domain adaptation to enhance generalizability.
- Actively researching the feasibility of large language models (LLMs) for disease diagnosis, reasoning, and treatment follow-up in medical applications.

Undergraduate Research Assistant

Department of Biomedical Engineering, KUET

February 2023 – September 2024

Khulna, Bangladesh

- Designed a hybrid model for gallbladder cancer classification using parallel transfer learning, feature fusion, and LSTM, achieving 99.37% accuracy; part of this work was published at ICEEICT 2024
- Gained experience in academic writing, literature review, and presenting research findings.

PUBLICATIONS (GOOGLE SCHOLAR)

Peer-Reviewed Journal Articles

Published:

- M. N. Hasan, **S. B. Shuvo**, M. M. H. Ankon, S. M. T. U. Raju, and N. Siddique, “**TransfusionNet: Framework for cervical cancer detection using deep learning with multi-level fusion**”, *Results in Engineering*, vol. 28, p. 107174, 2025, doi: 10.1016/j.rineng.2025.107174.

Under Review/In Preparation:

- M. N. Hasan, I. Ahmad, **S. B. Shuvo**, M. M. H. Ankon, S. Das, N. Siddique, and H. Wang, “**DL³M: A Vision-to-Language Framework for Expert-Level Medical Reasoning through Deep Learning and Large Language Models**,” arXiv preprint arXiv:2512.13742, 2025. Under review in *Medical Image Analysis*, Elsevier.
- **S. B. Shuvo**, S. A. Redhila, S. Hossain, and A. D. Roy, “**NeuroSwin: A Swin Transformer-GRU Model for Parkinson’s Disease Classification and Brain Region Localization from EEG Signals**,” Manuscript in preparation for *Biomedical Signal Processing and Control*, Elsevier.

- **S. B. Shuvo**, M. M. H. Ankon, M. A. Sayed, M. N. Hasan, and M. Z. Chowdhury, “**CholeNet: A Recurrent-Convolutional Hybrid Ensemble Framework with Quad-Fold Parallel Transfer Learning and Spatio-Sequential Feature Fusion for Gallbladder Cancer Classification**,” Manuscript in preparation for *Expert Systems with Applications, Elsevier*.
- M. M. H. Ankon, **S. B. Shuvo**, M. N. Hasan, and M. M. H. Manik, “**VRX-UNet: Automated Volumetric Characterization of Intra-tumoral Structures in Gliomas using 3D Segmentation**,” Manuscript in preparation for *Computers in Biology and Medicine, Elsevier*.
- M. A. Sayed, M. A. Rahman, **S. B. Shuvo**, M. J. Hossain, M. S. Alam, K. N. Hasan, and M. A. Kadir, “**Advanced image analysis for liver tumor detection and visualization in CT images using automated segmentation**,” Manuscript in preparation for *Nature - Scientific Reports*.
- M. N. Hasan, M. M. H. Ankon, **S. B. Shuvo**, I. Ahmad, M. M. H. Manik, and N. Siddique, “**Zero-to-Multi-Shot Benchmarking of Model-Agnostic Meta-Learning (MAML)-Based Domain Adaptation for Multi-Cancer Classification: Comparative Evaluation of LLMs, State-of-the-Art Deep Learning Models, and a Proposed Hybrid Approach**,” Ongoing Research.
- Md. N. Hasan, W. M. Shan, T. Joty, Md. M. H. Ankon, **S. B. Shuvo**, and N. Siddique, “**Unveiling Overfitting Patterns: Explainable AI-Driven Benchmarking of LLMs, Deep Learning Models, and a Hybrid Framework for Medical Image Classification**,” Ongoing Research.

Peer-Reviewed Conference Proceedings

Published:

- **S. B. Shuvo** and M. Z. Chowdhury, “**Classification of Gallbladder Cancer Using Average Ensemble Learning**,” *2024 6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT)*, Dhaka, Bangladesh, 2024, pp. 1450-1455, doi: 10.1109/ICEEICT62016.2024.10534480. [PDF].
- **S. B. Shuvo**, S. Hossain, S. A. Redhila, and U. Bose, “**Exploring Common Molecular Interactions across Multiple Cancers to Identify Potential Therapeutic Targets and Drug Candidates**,” *2025 7th International Conference on Electrical Information and Communication Technology (EICT)*, Dhaka, Bangladesh, 2025, pp. 1-6, doi: 10.1109/EICT68394.2025.11355627. [PDF]

Under Review:

- Md. A. Sayed, A. Rahman, **S. B. Shuvo**, M. A. Kadir, Md. I. A. Imran, Md. S. I. Wadud, and Md. N. Hossain, “**Automated PET to Fused PET-CT Mapping Using PCGAN for Lung Cancer Diagnosis**,” Submitted in *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking*, Chattogram, Bangladesh.

PROJECTS

- IoT Based Pilot Health Monitoring System** | *ESP8266 NodeMCU, Arduino, Blynk IoT* May 2018 – May 2020
- Developed an IoT-Based Pilot Health Monitoring System using ESP8266 NodeMCU, programmed with Arduino IDE, and sensors like MAX30100, DS18B20, and BMP180, integrated with the Blynk IoT platform to monitor pilot health and aircraft state, enhancing flight safety.
- MediConnect – A Video Conferencing App for Telemedicine and Healthcare** | *Java, Firebase* 2023
- Developed a secure video conferencing app for telemedicine as part of the Telemedicine & Health Care Laboratory individual project, utilizing Java, Firebase, and ZEGOCLOUD for effective remote healthcare communication.
 - Implemented features like sharing medical images, prescriptions, and real-time chat to support comprehensive virtual medical consultations.
- VitaSync: A Smart Band for Telemedicine Applications** | *Sensors, Mobile App Development* 2023
- Remodeled a wearable device for the Telemedicine & Health Care Laboratory group project, integrating sensors to track vital metrics such as heart rate, body temperature, and blood oxygen levels.
 - Developed a real-time data transmission system with a mobile app and an alert mechanism to notify users and contacts of health irregularities via email.
- Telechroma: A Smart Colorimetric Analyzer for Telemedicine Applications** | *Arduino, MIT App* 2022
- Developed Telechroma, a portable and affordable colorimetric analyzer for telemedicine applications, enabling remote diagnostics of glucose concentrations in urine samples.
 - Integrated hardware (Arduino Nano, TCS3200) with a mobile app for real-time data transmission and analysis.
- Sanitary Napkin Vending Machine** | *RFID, Arduino Mega, Stepper Motors* 2022
- Designed and built an automated Sanitary Napkin Vending Machine using RFID, Arduino Mega, stepper motors, and springs to promote menstrual hygiene through accessible and affordable technology.

TECHNICAL SKILLS

Programming Languages: MATLAB, Python (PyTorch, TensorFlow, Keras), R, C/C++
Simulation and Design Tools: NI Multisim, Arduino, Simulink, SOLIDWORKS, AutoCAD, COMSOL Multiphysics
Others: LaTeX, MS PowerPoint, MS Word, MS Excel, MS Visio, Adobe Photoshop, Adobe Illustrator, Adobe Premiere Pro

STANDARDIZED TEST SCORES

International English Language Testing System (IELTS)
Overall: 7.5 (Listening- 8.5, Reading- 7.0, Writing- 7.5, Speaking- 6.5)
Graduate Record Examination
Score: 313 (Quantitative 165, Verbal 148), AWA 3.5

16 November, 2024

1 October, 2024

TRAINING

Full day Technical Workshop on Biomaterials Based Wet Lab

Aug 2022

Bangladesh University of Engineering and Technology

Dhaka, Bangladesh

- Explored the work from the paper “Preparation of Different Polymorphs of Cellulose from Different Acid Hydrolysis Medium” (International Journal of Biological Macromolecules, 2019) and gained hands-on experience in preparing cellulose nanocrystals (CNC) using acid hydrolysis under expert guidance.
- Learned to operate lab equipment such as magnetic stir plates, centrifuges, ovens, autoclaves, FTIR, X-ray diffractometers, and thermogravimetric analyzers used for analyzing the morphology and properties of CNCs.

LEADERSHIP/EXTRACURRICULAR

Advanced Bioengineering Club of KUET (ABC-KUET)

Khulna, Bangladesh

President

March 2022 – Present

- Participated in multiple project-showcasing competitions and fostered innovation and collaboration among club members.
- Organized technical workshops, classes, and seminars, including “NeuroInquire: Exploring EEG and Research Foundations” and “ABC of Research with Hands-on Training on LaTeX.”

Fabrication Lab KUET

Khulna, Bangladesh

Student Operator

March 2022 – April 2024

- Assisted in organizing workshops such as ‘Hands-on Training Using Fabrication Tools’ and ‘IQAC’s Hands-on-Training Program’ at FabLab KUET, covering 3D printing, laser cutting, and CNC milling.
- Helped over 50 students fabricate project and thesis models, guiding them in transforming innovative ideas into real products.

Spectrum – Professional Skill Development Club of KUET

Khulna, Bangladesh

Head of Human Resources

March 2020 – March 2024

- Managed team operations, organized skill development events, and facilitated member engagement.
- Organized the Inter-University Case Study Competition, engaging 240 teams from 45 universities, securing sponsors like Bank Asia PLC, Abul Khair Group, Coca-Cola, and 10 Minute School, and distributing a prize pool of over BDT 100,000 while driving 300,000+ audience interactions.

AWARDS AND ACHIEVEMENTS

Honors in Bachelor of Science (B.Sc. Eng.)

February 2024

Dean’s Award (3 times)

Session: 2018-19, 2020-21, 2021-22

Technical Scholarship (4 consecutive times)

2019-2023

IEEE YESIST12 2023 – Finalist

2023

BME Indoor 2022 - Champion in Chess

2022

Intra-University Innovative Idea & Project Competition arranged by FabLab KUET - 2nd Runner-up

2022

National STEM Competition 2021 – Finalist

2021

National High School Programming Contest 2017 (Regional) - Position: 11th (Mymensingh Region)

2017

General Grade Scholarship in JSC Exam given by Board of Intermediate & Secondary Education, Dhaka

2013

Talent Pool Grade Scholarship in PSC Exam given by Board of Intermediate & Secondary Education, Dhaka

2010

CERTIFICATIONS

Neuroscience and Neuroimaging Specialization by Johns Hopkins University on Coursera

Ongoing

AI for Medicine Specialization by DeepLearning.AI on Coursera

Ongoing

Deep Learning Specialization by DeepLearning.AI on Coursera

Ongoing

Mathematics for Machine Learning Specialization by Imperial College London on Coursera

2025

Python 3 Programming Specialization by University of Michigan on Coursera

2025

Python for Everybody Specialization by University of Michigan on Coursera

2020

Introduction to Programming with MATLAB by Vanderbilt University on Coursera

2020

C for Everyone: Programming Fundamentals by UC Santa Cruz on Coursera

2020

Building Arduino robots and devices by Moscow Institute of Physics and Technology on Coursera

2020